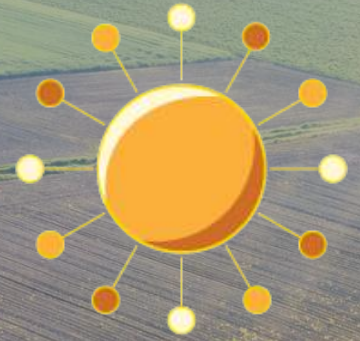


SolarSense



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What's the Problem?

Solar power is rapidly expanding across the nation. In the U.S. alone, “there over 235 gigawatts (GW) of solar capacity installed and a \$70.4 billion solar market.” In 2024, solar was above all other renewables like wind and hydropower, representing over 66% of new capacity” (SEIA).

Inefficient Solar Farm Layouts

Solar power is the future—but solar farms today aren’t as efficient as they could be. One of the biggest challenges solar developers face is a lack of tools for optimized solar farm layouts. Each panel must be placed with precision regarding weather patterns and geographical obstacles to best optimize the solar farm. Factors like uneven terrain, shading from trees and mountains, and incorrectly placed panel angles can reduce energy production by up to 10 percent. The poor placement of panels leads to wasted land, unnecessary costs, and lower energy output. This causes solar developers to spend more money to buy 30 to 50 acres plots of land that can cost upwards of \$500,000 just to hit their energy goals. According to a developer at Scout Clean Energy, “(Insert land quote, money is lost from inefficient layouts)

Economics + Health Issues

But why does inefficient solar technology matter to the average American? First of all, improving solar panel technology makes economic sense. Less sunlight captured means less energy produced and less money for solar developers and local communities. The 2024 global solar report stated that global revenue loss due to inefficient solar technology last year “reached \$4.6 billion, and that power loss due to equipment anomalies on solar farms rose from 3.13% to 4.47% year-over-year.” Inefficiencies within solar panel development cost billions and slows the development process. By improving solar panel technology, more energy is harnessed leading to a cleaner more sustainable future for everyone.

Improvements in solar technology not only saves money but also saves lives. In 2024, over “39% of people living in America—131.2 million people—still live in places with failing grades for unhealthy levels of ozone or particle pollution.” (lung). In Alabama, “industrial facilities released more than 82 million pounds of toxic material into the air, land and water leading to an increase in serious health effects such as asthma attacks, cardiovascular damage, and developmental issues,” (lung).

Is there a way to find cleaner energy sources that can mitigate rising pollution levels and negative health effects? The answer lies in developing renewable energy sources like solar power. According to the Department of Energy, renewable energy is making the air and water in our communities cleaner, improving health outcomes for vulnerable groups, like children, and saving Americans up to \$77 billion per year in total health costs.” Investing in solar technology is more than addressing inefficiencies, it’s improving the lives of everyday Americans, one solar panel at a time.

The problem is clear: inefficiencies in solar farm layouts leads to a loss of energy and money. So how can we better harness the power of the sun and improve solar technology? Imagine a software that shows you the perfect layout of solar panels on any piece of land to optimize the energy potential, cutting wasted land costs and improving efficiency. This software leads to more captured energy contributing to a cleaner and more sustainable future for everyone. That’s exactly what our software, SolarSense, will do.

Our Solution

Overview

Introducing SolarSense, an easy and intuitive platform where solar farm developers can design and optimize the best layouts for solar farms.

SolarSense is an innovative software solution that leverages machine learning and satellite data to optimize solar farm layouts and design. This platform enables solar developers to minimize land and material costs and maximize energy efficiency by analyzing key environmental and infrastructural factors such as weather patterns, cloud cover, transmission lines, and geographical obstructions. Through a user-friendly interface, SolarSense provides real-time, data-driven recommendations for solar panel placement, making the design process faster, more efficient, and highly precise. Our machine-learning software adapts its recommendations as new data becomes available to offer the best possible solar farm layout.



SolarSense software enhances a solar farm’s layout, reducing required land for a solar farm.

“Hot Zones”

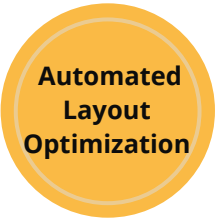
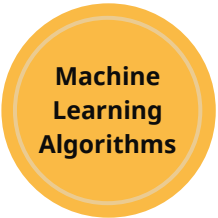
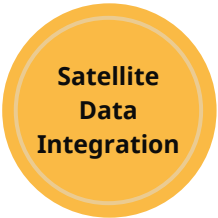


Additionally, our “hot zones” feature is a powerful tool for solar developers, enabling them to analyze large plots of land and assess their potential for optimal energy capture. This can play a crucial part in determining the best plot of land to build the most efficient solar farm possible.

As shown in the graphic above, the green zones represent areas of the highest efficiency and energy yield because of large open areas with direct sunlight, good weather conditions, and minimal natural obstructions. This indicates the most optimal zones to place these solar farms. However, red zones represent areas of the lowest efficiency due to geographical obstacles such as trees, hilly terrain, or typical severe weather conditions. Finally, the yellow zones represent moderate energy yields due to factors such as partially shaded areas and average weather conditions.

The picture above is just a model, but in the final version the hot zones would have a larger gradient with extended information. This tool of our software is used for initial analysis of the land area.

Step by Step: How it works



1. Initial Setup and Area Selection
 - Solar developers start by accessing SolarSense’s user-friendly website, which features an interactive 3D landscape of the selected area.
 - They can zoom in and out of the map to view different regions and select the specific land section they want to assess for solar farm potential.
2. Satellite Data Integration
 - SolarSense automatically collects high-resolution satellite imagery of the selected area.
 - The platform analyzes land topography, shade zones, and potential obstacles such as trees, buildings, or uneven terrain
 - The data is then processed to identify the most suitable areas for solar panel placement, marked clearly with color-coded “hot zones” to show regions with the highest and lowest solar potential.
3. Machine Learning Algorithms
 - After the land plot is selected, the platform uses machine learning algorithms to process weather condition data and cloud cover patterns. It will also overlay regional electrical grid maps onto the 3D landscape, showing the proximity of the land to nearby transmission lines.
 - This information helps developers assess the feasibility of connecting the solar farm to the grid and optimizes energy distribution to minimize infrastructure costs.
 - The platform also integrates local solar policies, permitting requirements, and incentives, ensuring that the solar farm design complies with regulatory standards and benefits from available incentives.
4. Automated Layout Optimization
 - After processing, the platform generates a fully optimized solar farm layout based on the data-driven recommendations.
 - The automated system reduces planning time by offering efficient, high-quality design suggestions with minimal user input.
 - The final layout is tailored to maximize energy generation, minimize infrastructure costs, and ensure compliance with grid connectivity and local policies.

What’s the Benefit?

How does SolarSense tackle inefficiencies within solar farm layouts? SolarSense provides the best-designed solar farm layout from the start, reducing land usage, cutting costs and improving energy generation. Because of inefficient layouts, it takes 10 acres of land just to produce 1 MW. Not including the cost of the land, this solar farm would cost about \$1 million to build. Now as previously stated, solar developers currently are forced to buy plots of land ranging from 30 to 40 acres and sometimes even 200 acres just to meet the minimum energy quota. The cost of these solar farms can be upwards of \$2 billion to build. However with SolarSense, more optimized layouts produce the same energy in up to 50% less acreage. Or, if developers still choose to develop solar farms in a 200-acre area, SolarSense’s optimized layout would increase their energy production, boosting revenue.

We’ve already been in contact with over 40 solar developers across the nation in order to better tailor our software for industry needs. One developer from Scout Clean Energy provided supportive feedback for SolarSense saying, “Wonderful idea! The solar industry would really benefit from technology like this!”

Expected Challenges



CHALLENGE

Our main challenge with SolarSense is fully developing its highly complex software. SolarSense analyzes an extreme volume of data across multiple different sets to provide optimal solar farm layouts.

SOLUTION

We plan to work with a software developer to fully develop the software. SolarSense uses machine learning incorporated in its programming to help filter and analyze these large quantities of data for farm layouts.

CHALLENGE

Our next major challenge is insert ourselves into a competitive solar market. The market is constantly growing and changing which means SolarSense must be ready to adapt.

SOLUTION

We will partner with NextTracker, an American solar tracker developer and manufacturer that already has a secure market and the most optimized solar panel design. This will allow us the best access to the solar developer market and to refine our software as the market changes.

CHALLENGE

Another challenge is securing sufficient funding to fully develop our high-tech software. Because of our budgetary needs, it may take longer than expected to secure enough funding on our timeline.

SOLUTION

We will seek investor funding through business pitch competitions such as Big Ideas and ALDAG locally and then move on to a national level through the Department of Energy American Made Solar Prize Competition.

COMPETITORS

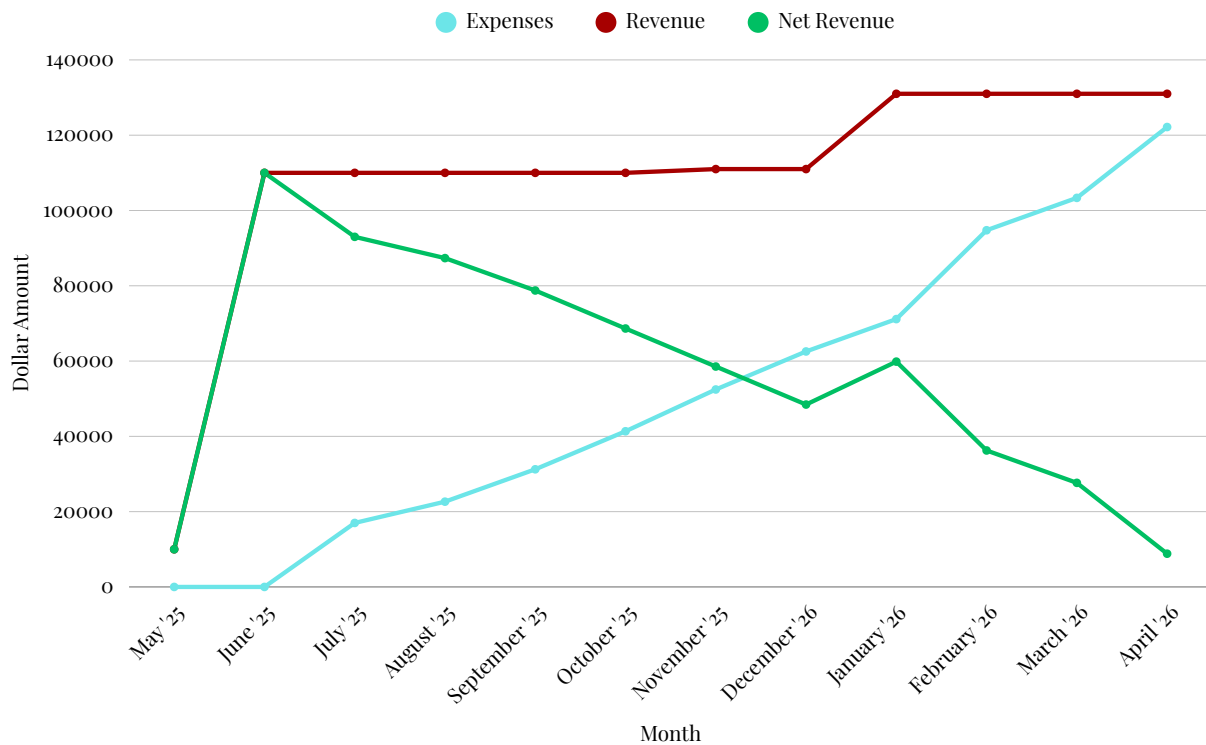
Traits	SOLARSENSE	PV CASE	HELISCOPE	SOLAREEDGE
Solar Farm Design	✓	✓	✓	✗
User Friendly Interface	✓	✗	✓	✗
Best layout prediction	✓	✗	✗	✗
Large scale solar farms	✓	✓	✓	✗
Machine Learning	✓	✗	✗	✗
Real Time Weather Integration	✓	✗	✗	✓
3D Visualization	✓	✗	✓	✗

How we're Different:

SolarSense differentiates itself from competitors like PVCase, Helioscope and SolarEDGE by offering a more integrated, user-friendly approach that combines machine learning, satellite and weather condition data, and real-time adaptability into one seamless platform. While tools like PVCase and Helioscope provide advanced solar design and layout features, SolarSense goes a step further by leveraging high-resolution satellite imagery and continuously processing environmental data, including weather patterns and cloud cover, to provide dynamic, real-time optimization of solar farm layouts. Unlike PVCase, which focuses heavily on specific design details, or Helioscope, which offers performance modeling, SolarSense incorporates a comprehensive “hot zones” feature that provides a broad, data-driven overview of the entire land area, highlighting regions with the highest and lowest solar potential based on environmental factors.

Additionally, SolarSense's automated layout optimization system significantly reduces the manual effort required in site planning. This feature, combined with its ability to integrate regional electrical grid maps and local policy data, streamlines the entire process, making it easier for solar developers to design cost-effective, high-efficiency solar farms without needing deep technical expertise. This combination of real-time recommendations, user-friendly interface, and powerful data analysis tools sets SolarSense apart as a comprehensive solution for solar developers.

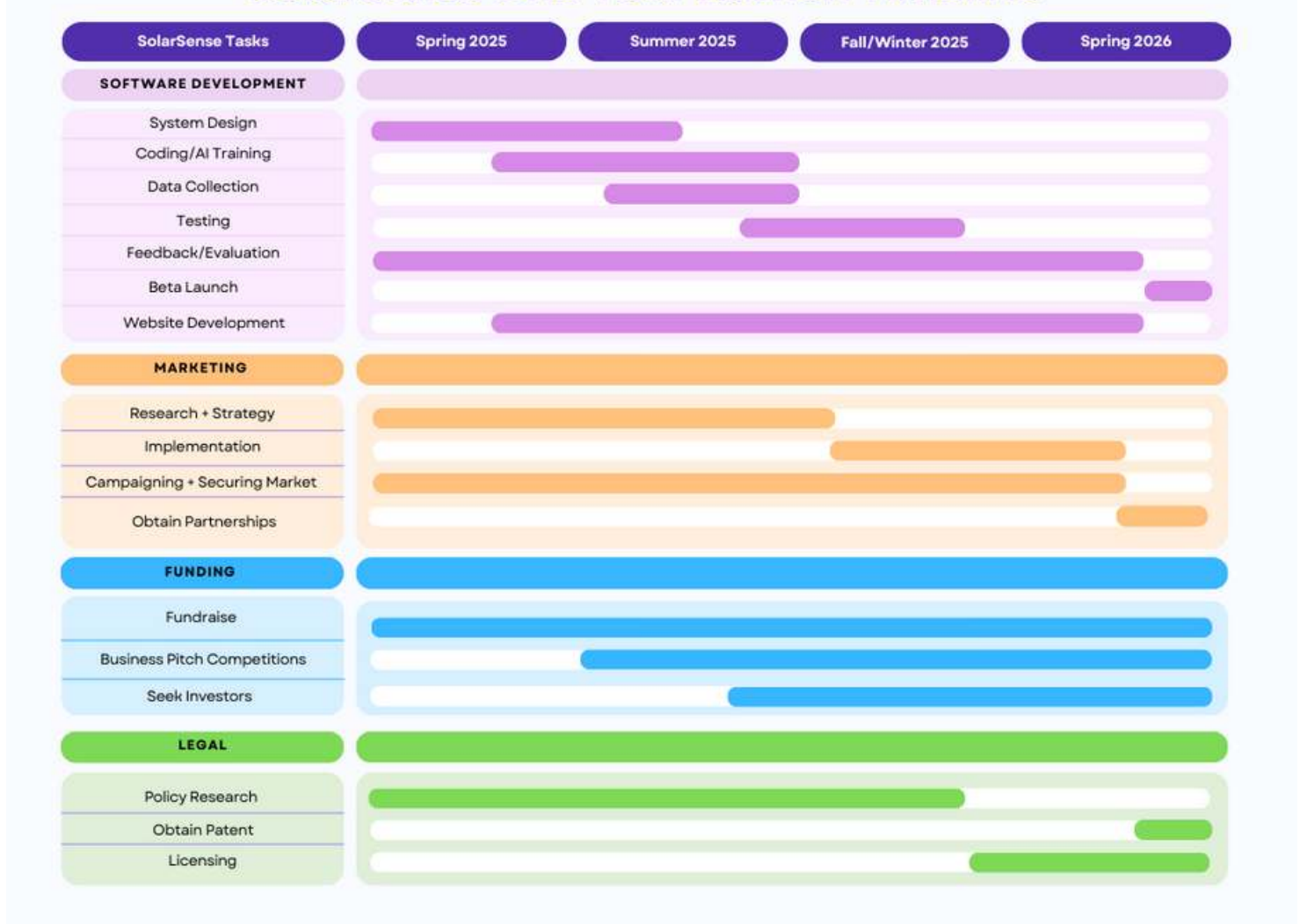
Financial Report



Solar energy is a very expensive industry to try and launch into. Because what we are working on is very high scale and complex, our software development, data collection, and implementation all have extreme costs. However, we believe that the funding for this idea is possible through crucial steps. First, we plan to participate in multiple business pitch competitions, both local and on the national circuit. The main competitions include: Big Ideas, ALDAG, and the Department of Energy American Made Solar Prize Competition. While we hope to place in each competition, our main goal is to gain awareness in order to attract investors to gain funding. Second, once we create a base prototype of SolarSense with our investor and competition winnings, we will apply for solar related grants specifically in the state of Texas. Finally, we hope to continue to gain awareness and funding in the solar market to make SolarSense possible.

Once our model is fully developed, we will partner with NEXTracker to continue to expand SolarSense and reach more of our market. SolarSense will be a Software-as-a-Service platform and will be offered as a yearly subscription to solar developers. This will be our main source of revenue but we will have a smaller revenue stream through advertisements on our software. Our main costs will include continually accessing high quality, private weather and satellite data, paying a software developer to maintain our interface, and paying a marketing agency to continuously spread SolarSense across the solar market.

SolarSense One Year Launch Timeline



This is our planned timeline that is closely integrated to our financial plan. We want to be as aggressive as possible with SolarSense’s development yet realistic. We have broken down the timeline for the next year into four different categories. Starting with software development, we will begin system design and interface as soon as possible. Next, we will start coding and simultaneously developing SolarSense’s website. Finally, after several rounds of testing and prototype development, we will beta launch SolarSense in the spring of 2026.

Marketing also plays a major role in SolarSense. We must gain awareness for our company as soon as possible because of the competitive nature of this market. This includes campaigning alongside researching and developing a market strategy. This will allow us to secure the market and obtain the required partnerships with companies such as NEXTracker.

Funding is the most critical part of this entire timeline. We need to begin fundraising immediately in order to successfully develop our software and launch SolarSense into the solar market. We will fundraise until our software model is fully complete, where SolarSense will receive steady revenue.

Finally, we will research all policies and patent information required to officially launch SolarSense and secure the ability for SolarSense to remain a step above the competition for years to come.



Measuring Success

Meet Our Team



MORGAN BYERLEY BATON ROGUE, LOUISIANA

Morgan Byerley is a senior at the University of Alabama studying international studies with minors in German and English. She has studied abroad in two countries, published over fifteen articles, and created hundreds of social media posts for nonprofits. Currently she is working as an intern for a clean energy nonprofit where she markets and educates about the transition to clean energy within Alabama. While studying abroad in Seoul, South Korea and Berlin, Germany, Morgan worked on an ESG marketing project and studied green business practices in startups. Observing and studying how other countries had progressed in clean energy and sustainable practices gave her a new perspective on her home state of Alabama.



TOMAS BALTZER COLORADO SPRINGS, COLORADO

A junior Mechanical Engineering student at The University of Alabama and part of the STEM-to-MBA Honors Program, Tomas brings a background of both engineering and business. Tomas has gained hands-on experience through prestigious internships, including contributing to a \$20M plasma energy research initiative and optimizing sealer delivery at Mercedes-Benz, projecting a total of \$1M+ in savings. For his business experience, Tomas participated and won first prize in the University's IP Commercialization Academy, building a business around patented energy-efficient technology as well as giving multiple pitch presentations through the STEM/MBA program.



EMILY RABBIDEAU BIRMINGHAM, ALABAMA

Emily is a visual storyteller, bringing narratives to life through a wide variety of mediums. As a senior at the University of Alabama, she is pursuing a double major in Studio Art and Creative Media with a minor in Spanish. Emily has a background in documentary production, having produced videos for brands such as CNBC and Mercedes-Benz in addition to an Emmy-award winning production house at the Capstone. Her work typically deals with niche communities and technical topics, making complex issues accessible to a generic audience. Emily strives to approach all aspects of her life creatively, bringing a unique way of thinking to a team.

Sources